

1.INTRODUCTION

1.1 Project overview:

Auto Ticket Classification using Flow Designer is a ServiceNow-based automation project designed to automatically classify and route support tickets. When a user creates a ticket, the system analyses the issue details and identifies the appropriate category and subcategory based on predefined conditions.

Using Flow Designer, the system can automatically assign the ticket to the appropriate support group and update the required ticket fields. This reduces manual classification, minimizes errors, saves time, and improves the overall efficiency of ticket management.

1.2 Purpose of the Project:

The main purpose of **Auto Ticket Classification using Flow Designer** is to automate the classification and assignment of support tickets.

- Automatically identify the type of issue from the ticket details.
- Assign the appropriate **category and subcategory**.
- Route the ticket to the correct **support or assignment group**.
- Reduce manual effort and human errors.
- Improve the speed and efficiency of ticket management.
- Provide a more organized and consistent ticket-handling process.

2.IDEATION PHASE

2.1 Problem Statement:

In a traditional ticket management system, support tickets are often **manually reviewed, categorized, and assigned** to the appropriate support team. This process can be time-consuming and may lead to **incorrect classification, delays, and human errors**, especially when the number of tickets increases.

Therefore, there is a need for an automated solution that can **identify the type of issue, classify the ticket, and assign it to the appropriate support group** without requiring manual intervention.

The proposed project, “**Auto Ticket Classification using Flow Designer,**” aims to address this problem by using ServiceNow Flow Designer to automate the ticket classification and assignment process, improving the **speed, accuracy, and efficiency** of ticket management.

2.2 Empathy Map Canvas

An **Empathy Map Canvas** helps us understand the **user's needs, problems, thoughts, and expectations** while using the ticket management system.

Says

“I need to raise a ticket quickly.”

“My issue is urgent.”

“I don't know which category to select.”

Thinks

“Will my ticket reach the correct team?”

“How long will it take to resolve?”

“I hope I entered the issue correctly.”

Does

Creates a support ticket.

Enters the issue description.

Checks the ticket status.

Waits for support team response. Satisfied when the issue is handled efficiently.

Feels

Frustrated when ticket processing is delayed.

Confused about selecting the right category.

Relieved when the ticket is assigned quickly.

Pain Points

- Manual ticket classification takes time.
- Users may select the wrong category.
- Tickets may be assigned to the wrong team.
- Delays can occur in ticket processing.

- Support staff have to handle repetitive classification tasks.

Needs / Gains

- Quick ticket creation.
- Automatic and accurate classification.
- Correct assignment to the support team.
- Faster ticket processing.
- Reduced manual work.
- Easy tracking of ticket status.

2.3 Brainstorming

Brainstorming is the process of generating different ideas and possible solutions for a problem before selecting the most suitable approach.

Brainstormed Ideas

1. **Automatic Category Assignment**
Identify the ticket category based on the issue description.
2. **Keyword-Based Classification**
Use keywords such as *Wi-Fi*, *password*, *laptop*, *software*, etc. to identify the issue type.
3. **Automatic Assignment Group**
Route the ticket to the appropriate team based on its category.
4. **Automatic Priority Assignment**
Set ticket priority based on predefined conditions such as urgency or issue type.
5. **Automatic Notifications**
Notify the assigned support team when a new ticket is created.
6. **Status Automation**
Automatically update the ticket status when specific actions are completed.

7. Duplicate Ticket Detection

Identify similar or duplicate tickets to avoid unnecessary work.

8. Ticket Escalation

Automatically escalate unresolved or high-priority tickets after a specified time.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	User Experience	Problem / Pain Point	System Response
1. Identify Issue	User faces a technical problem	Confused / concerned	Doesn't know how to solve the issue	Provides a ticketing system
2. Create Ticket	Enters issue details and submits ticket	Hopeful	May not know the correct category	Accepts ticket details
3. Auto Classification	Waits for processing	Relieved	Manual classification can cause delays	Flow Designer automatically identifies the category
4. Ticket Assignment	Waits for support	Confident	Ticket could be assigned to the wrong team	Automatically assigns the correct support group

Stage	User Action	User Experience	Problem / Pain Point	System Response
5. Ticket Resolution	Communicates with support team	Satisfied	Delayed responses may affect satisfaction	Support team works on the assigned ticket
6. Closure	Confirms issue is resolved	Satisfied	—	Ticket status is updated and closed

3.2 Solution Requirements

The **Solution Requirements** define what the proposed system should provide to solve the problems identified in the ticket management process.

1. Functional Requirements

- **Ticket Creation:** Users should be able to create a ticket by entering the issue details.
- **Automatic Classification:** The system should identify the appropriate category based on the ticket information.
- **Subcategory Assignment:** The system should assign a suitable subcategory where applicable.
- **Automatic Assignment:** Tickets should be automatically assigned to the appropriate support group.
- **Priority Management:** The system should assign priority based on predefined conditions.
- **Status Management:** The ticket status should be updated throughout the workflow.
- **Notifications:** Relevant users or support teams should receive notifications when tickets are created or assigned.

- **Ticket Tracking:** Users should be able to view and track their ticket status.

2. Non-Functional Requirements

- **Usability:** The system should have a simple and user-friendly interface.
- **Performance:** Ticket classification and assignment should happen quickly.
- **Reliability:** The flow should work consistently for every valid ticket.
- **Scalability:** The system should be able to handle an increasing number of tickets.
- **Security:** Only authorized users should be able to access or modify ticket information.
- **Maintainability:** Classification rules and workflow conditions should be easy to update.

3. Technical Requirements

- **Platform:** ServiceNow
- **Automation Tool:** Flow Designer
- **Database:** Custom Ticket Table
- **Interface:** ServiceNow Form / Portal
- **Automation:** Triggers, Conditions, and Actions
- **Notification:** ServiceNow notification mechanism

Overall Requirement

The system should automatically analyze newly created tickets, classify them according to predefined rules, assign them to the appropriate support group, and manage the ticket workflow with minimum manual intervention.

3.3 Data Flow Diagram (DFD)

- ◆ Level 0 DFD – Context Diagram

USER



Ticket Details



AUTO TICKET
CLASSIFICATION SYSTEM
(Flow Designer)



Classified Ticket



SUPPORT TEAM



Ticket Status



USER

3.4 Technology Stack

Technology	Purpose
ServiceNow	Main platform used to build the ticket management system
Flow Designer	Automates ticket classification, assignment, and workflow
Custom Ticket Table	Stores ticket-related information
Form Designer	Creates and organizes the ticket submission form
ServiceNow UI	Provides the interface for users and support staff
Business Rules / Conditions	Defines rules for ticket classification
Notifications	Sends alerts to users or support teams
ServiceNow Database	Stores ticket details, categories, assignments, and status

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Problem	Proposed Solution	Expected Benefit
Manual ticket classification	Automate classification using Flow Designer	Reduces manual effort
Incorrect category selection	Apply predefined classification rules	Improves consistency
Tickets assigned to the wrong team	Automatically select the appropriate assignment group	Better ticket routing
Delays in ticket processing	Trigger automated workflows immediately after ticket creation	Faster processing

Problem	Proposed Solution	Expected Benefit
Repetitive work for support staff	Automate routine classification and assignment tasks	Reduces workload
Difficulty tracking tickets	Maintain ticket status in the system	Better visibility
Delayed communication	Use automated notifications	Faster communication

4.2 Proposed Solution

The proposed solution is to develop an **automated ticket classification and assignment system using ServiceNow Flow Designer**. The system will reduce manual effort by automatically analyzing newly created tickets and routing them to the appropriate support team.

◆ How the Proposed Solution Works

1. Ticket Creation

The user enters the issue details through the ticket form.

2. Flow Trigger

When a new ticket is created, the Flow Designer is automatically triggered.

3. Ticket Analysis

The system checks the ticket's short description and other relevant information.

4. Automatic Classification

Predefined conditions are used to identify the appropriate category and subcategory.

5. Automatic Assignment

Based on the classification, the ticket is assigned to the relevant support group.

6. Priority & Status Update

The system can automatically set priority and update the ticket status according to predefined rules.

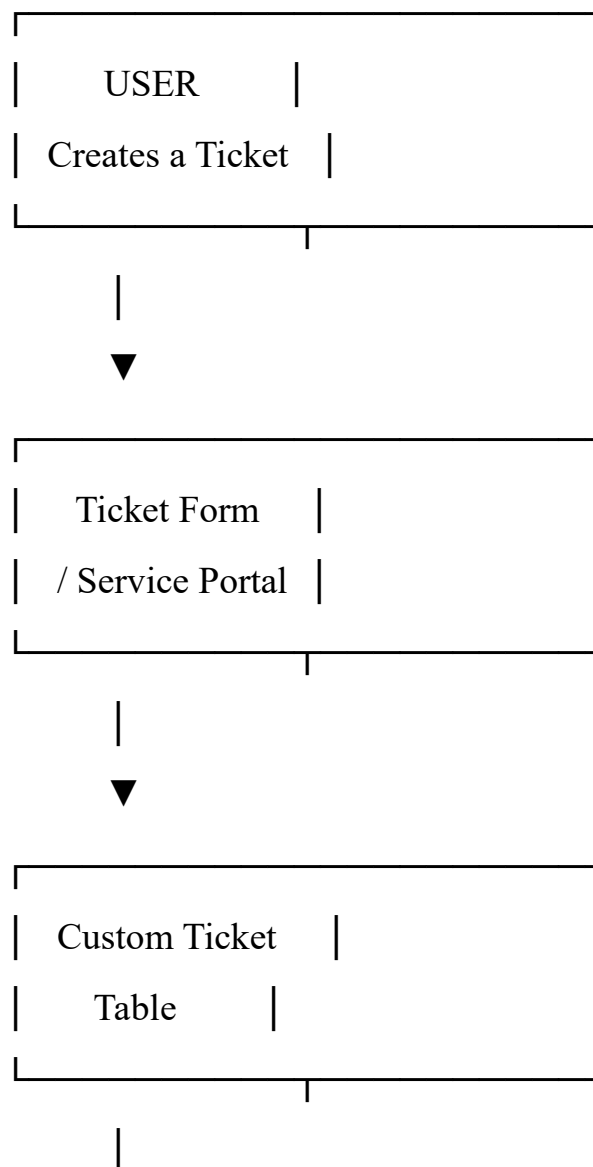
7. Notification

The assigned support team can receive an automatic notification.

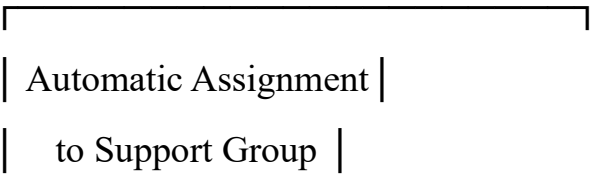
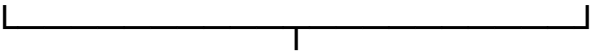
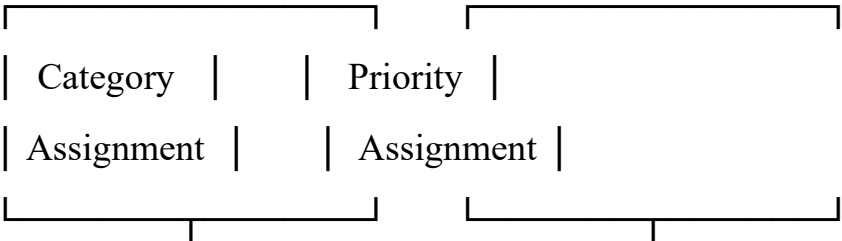
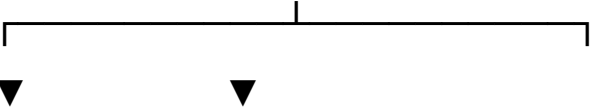
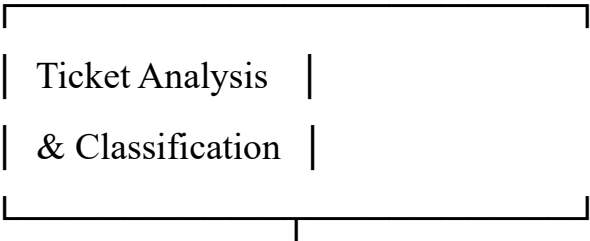
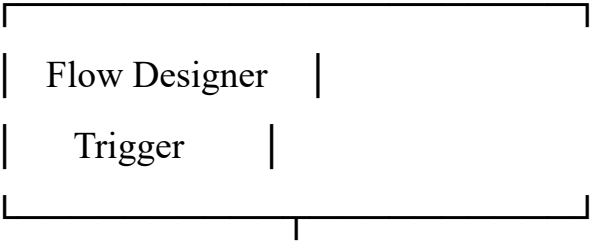
8. Ticket Resolution

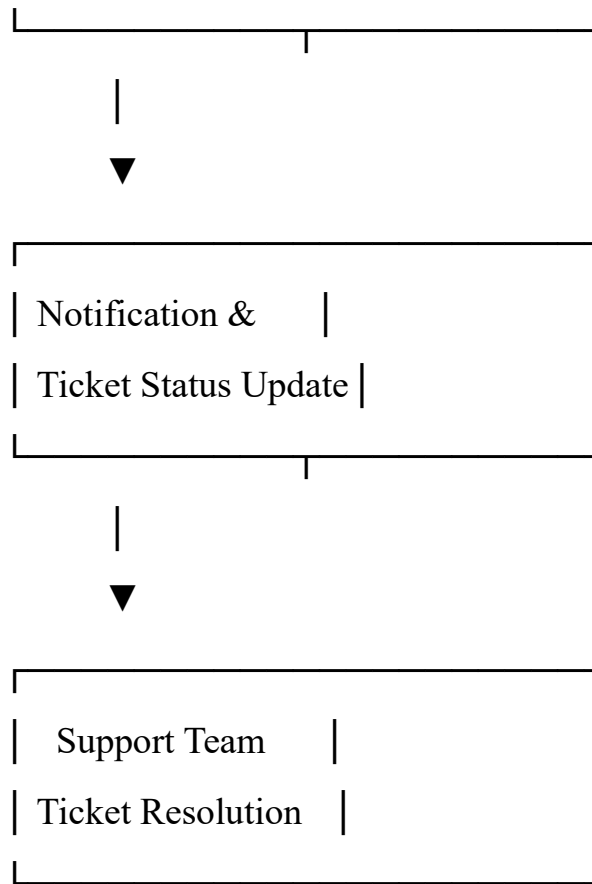
The support team handles the issue and updates the ticket until it is resolved and closed.

4.3 Solution Architecture



Ticket Created





5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Expected Output
1. Requirement Analysis	Identify ticket management problems and collect requirements	Project requirements
2. System Design	Design ticket fields, workflow, DFD, and classification rules	System design
3. Table Creation	Create the custom ticket table and required fields	Ticket database
4. Form Design	Design the ticket creation and management forms	User-friendly ticket form

Phase	Activities	Expected Output
5. Flow Development	Create triggers, conditions, and actions in Flow Designer	Automated workflow
6. Classification Setup	Define rules for categories and subcategories	Auto-classification
7. Assignment Setup	Configure support groups and automatic routing	Auto-assignment
8. Notification Setup	Configure notifications for ticket creation and assignment	Automated alerts
9. Testing	Test different types of tickets and workflow conditions	Tested system
10. Deployment & Documentation	Finalize the system and prepare project documentation	Completed project

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance Testing Flow

Create Test Tickets



Flow Designer Triggered



Measure Classification Time



Check Category & Assignment



Check Processing Errors



Analyse Results

Performance Metrics

You can measure:

- **Response Time** – How quickly the ticket is processed.
- **Processing Time** – Time taken for classification and assignment.
- **Success Rate** – Percentage of tickets correctly processed.
- **Error Rate** – Number of tickets that fail or are incorrectly processed.
- **Throughput** – Number of tickets processed within a given period.

7. RESULTS

Dashboard > Skillwallet > Program > Servicenow system administrato... > Group-projects > Auto Ticket CI

0 AI Token300 CreditsAP

Auto Ticket Classification using Flow Designer

TeamOverviewWorkspaceKanban

Milestone 1: Requirement Analysis & Planning

EpicComplexity: MediumDuration: 15m

Description

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues. Currently, IT staff manually review each ticket to:

- Identify the issue type.
- Select the correct category and subcategory.
- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

Resources

No resources available

Project Progress90%

View Demo

View Repo

Milestone 1: Requirement Analysis & Planning90%

Activity 1: Business Use Case

Activity 2: Creation of New Update Set

Activity 3: Navigation steps in Picture View

Milestone 2: Backend Development & Configuration90%

Milestone 3: Automation using Flow Designer & Email Notification90%

Milestone 4: Testing, Validation & Security90%

Milestone 5: Deployment & Conclusion90%

8. ADVANTAGES & DISADVANTAGES

Advantages & Disadvantages

✓ Advantages

1. **Reduces Manual Work**

Automatically classifies and assigns tickets without manual intervention.

2. **Saves Time**

Tickets can be categorized and routed immediately after creation.

3. **Reduces Human Errors**

Predefined rules help maintain consistent classification.

4. **Faster Ticket Assignment**

Tickets are automatically sent to the appropriate support group.

5. **Improves Efficiency**

Support teams can focus more on solving issues rather than sorting tickets.

6. **Easy to Track**

Ticket category, assignment, and status can be managed in one system.

7. **Scalable**

The workflow can handle increasing numbers of tickets with appropriate configuration.

✗ Disadvantages

1. **Rule Dependency**

The system depends on predefined conditions and keywords.

Unrecognized issues may not be classified correctly.

2. **Initial Setup Required**

Creating tables, forms, flows, conditions, and assignment rules requires time and configuration.

3. **Maintenance Required**

Classification rules may need to be updated when new types of issues are introduced.

4. **Complex Issues**

Tickets containing multiple or unclear issues may be difficult to classify automatically.

5. **Platform Dependency**

The solution depends on the ServiceNow platform and its available features.

6. **Configuration Errors**

Incorrect flow conditions can result in wrong categorization or assignment.

9. CONCLUSION

The **Auto Ticket Classification using Flow Designer** project provides an automated approach to managing and processing support tickets. By using **ServiceNow Flow Designer**, tickets can be automatically analyzed, classified, and assigned to the appropriate support group based on predefined conditions.

This solution helps **reduce manual effort, minimize classification errors, save processing time, and improve ticket management efficiency**. It also provides a structured workflow for tracking and resolving tickets.

Overall, the project demonstrates how **workflow automation can simplify IT support processes and provide a more efficient and organized ticket management system**.

10.FUTURE SCOPE

☐ **AI-Based Classification**

Machine Learning or AI can be used to understand ticket descriptions and classify tickets more intelligently.

☐ **Natural Language Processing (NLP)**

NLP can help the system understand different ways users describe the same issue.

☐ **Automatic Priority Prediction**

The system can analyze the ticket content and automatically determine its priority.

☐ **Sentiment Analysis**

User messages can be analyzed to identify the level of urgency or frustration expressed in the ticket.

☐ **Duplicate Ticket Detection**

The system can identify similar existing tickets and avoid creating unnecessary duplicate tickets.

☐ **Automatic Response Generation**

AI can generate an initial response or suggest troubleshooting steps to the user.

☐ **Predictive Issue Resolution**

Historical ticket data can be analyzed to suggest possible solutions for new issues.

☐ **Advanced Analytics Dashboard**

Dashboards can provide insights into ticket volume, categories, resolution time, and support-team performance.